

com/2021/03/the-latest-trends-in-the-programming-world/. The same criteria were used to compile the Top 10 list, and the programming languages were: C, C#, C++, Java, JavaScript, PHP, Python, R, Swift, and TypeScript. In this edition, Objective-C was replaced by TypeScript. Notably, eight languages consistently appeared in the Top 10 across all three of my rankings from 2017 onwards: C, C#, C++, Java, JavaScript, PHP, Python, and Swift. This article also identified the promising programming languages. In addition to Python and JavaScript, which were highlighted in 2019, R and TypeScript showed strong promise for rapid adoption in the programming world. Like the earlier edition, a list of languages in decline was compiled, featuring Perl (also present in the 2019 list) and Objective-C, whereas Ruby showed improvement and was removed from the list.

The list of promising young programming languages in 2021 remained largely the same as in 2019, with the addition of an eighth language, Solidity, which is used for blockchain-based applications. The final list therefore included Swift, TypeScript, Go, Rust, Kotlin, Dart, Julia, and Solidity.

Now that five years have passed since the last article with this theme was published, it is time to analyse the latest trends in the programming world. Our analysis will be based on the same four rankings used in the previous three articles mentioned earlier. This article is particularly relevant in light of two major developments since the last edition of the Top 10 programming languages: the emergence of ChatGPT, Gemini, and other large language models (LLMs), and significant research breakthroughs in quantum computing. It is high time to analyse how these impactful technologies have affected the world of programming.

The Top 10 list

We now examine how recent developments in computer science

have shaped trends in the programming world. As a starting point, we present a Top 10 list of programming languages. The ranking in this article is obtained by aggregating the latest results from the four rankings introduced earlier. Drawing on multiple indices helps mitigate the biases inherent in any single ranking and offers a more balanced assessment. Six languages—Python, Java, JavaScript, C, C++, and C#—appear consistently across all four ranking schemes. These languages were also permanent fixtures in our previous rankings and continue to retain their positions in 2026, underscoring their enduring relevance and influence in modern computing.

The next two languages included in our Top 10 list are PHP and R. Each has recorded two Top 10 finishes and four Top 20 finishes across the rankings we considered. PHP had consistently appeared in the Top 10 in our previous three editions; however, its current positions—12th in the IEEE Spectrum programming languages ranking and 15th in the TIOBE index—suggest a decline in its usage over the past five years. R, by contrast, ranks 12th in the RedMonk ranking and 11th in the IEEE Spectrum ranking, placing it slightly ahead of PHP in overall popularity.

However, the final two spots in the Top 10 list are closely contested among TypeScript (2 Top 10 finishes and 3 Top 20 finishes), Rust (1 Top 10 finish and 4 Top 20 finishes), Swift (1 Top 10 finish and 3 Top 20 finishes), Go (1 Top 10 finish and 3 Top 20 finishes), and Kotlin (4 Top 20 finishes). To resolve this tie, I selected the two languages with the best average ranking across the four rankings considered: RedMonk, PYPL, IEEE Spectrum, and the TIOBE index. Rust is ranked 19, 7, 14, and 13 respectively, yielding the strongest average among the candidates at 13.25. Swift follows with rankings of 11, 6, 16, and 22, resulting in an average of 13.75. The remaining contenders trail slightly behind: TypeScript averages 14.25 (6,

12, 7, 32), Go averages 14.75 (12, 21, 10, 16), and Kotlin averages 16 (14, 17, 13, 20). Based on these aggregate rankings across independent measures, Rust and Swift emerge as the most consistent performers and therefore complete the final two positions in the Top 10 list.

In the previous three rankings, we only considered whether a language appeared in the top 10 or not, without assigning a numeric order within that group. In this article, however, we apply the same averaging method used earlier to break ties and produce a relative popularity ranking of the ten selected languages, arranging them from most to least popular. Thus, the final ranking based on average positions across the four indices is as follows:

1. Python
2. Java
3. C++
4. JavaScript
5. C
6. C#
7. R
8. PHP
9. Rust
10. Swift

Before drawing conclusions from the inclusions and omissions in the list, it is useful to compare the four ranking schemes and the biases inherent in each. We begin with the RedMonk ranking, which reports only the top 20 programming languages. Its methodology relies on two parameters: GitHub activity and Stack Overflow discussions. As a result, this ranking serves as a strong indicator of developer engagement and community activity. However, it tends to favour programming languages with vibrant open source ecosystems while under-representing programming languages used primarily in closed enterprise environments. Notably, this is the only ranking in which Python is placed second, narrowly trailing JavaScript.